

P₁ 习题 6 P₂

I) $xy' = 2y \quad y = 5x^2$

$$y' = 10x \quad 10x^2 = 10x^2 \rightarrow \text{是}$$

II). $(y')^2 + 4y^2 - 4 = 0$

$$y' = 2\cos(2x+C)$$

$$4\cos^2(2x+C) + 4\sin^2(2x+C) - 4 = 4 - 4 = 0 \rightarrow \text{是}$$

III). $y'' + 4y' + 4y = 0$

$$y_1' = e^{-2x} - 2xe^{-2x} \quad y_1'' = -2e^{-2x} - 2e^{-2x} + 4xe^{-2x}$$

$$-2e^{-2x} - 2e^{-2x} + 4xe^{-2x} + 4e^{-2x} - 8xe^{-2x} + 4xe^{-2x} = 0 \rightarrow y_1 \text{ 是}$$

$$y_2' = y_2'' = y_2 = e^x$$

$$e^x + 4e^x + 4e^x \neq 0 \rightarrow y_2 \text{ 不是}$$

IV) $(1+xy)y' + y^2 = 0$

$$x'(t) = e^t + te^t \quad y'(t) = -e^{-t}$$

$$y' = \frac{y'(t)}{x'(t)} = -\frac{e^{-t}}{e^t + te^t} = -\frac{1}{e^{2t}(t+1)}$$

$$-(1+t)\frac{1}{e^{2t}(t+1)} + e^{-2t} = \frac{-(1+t) + (t+1)}{e^{2t}(t^2+1)} = 0 \rightarrow \text{是}$$

P₂ 习题 6 P₃

$$y'' + 2y' + 3y = e^{3x}$$

$$x=0, \quad y'|_{x=0} = e^0 = 1$$

$$\left(\frac{2x}{1+x^2}\right)'$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{f(x)} = \lim_{x \rightarrow 0} \frac{\frac{2x}{1+x^2}}{f'(x)} = \lim_{x \rightarrow 0} \frac{2(1+x^2) - 4x^2}{(1+x^2)^2} = \frac{2}{1} = 2$$

P₃ 习题 6 P₄.

I). $x^2 y' + y = 0$

$$x^2 y' = -y \quad \frac{dy}{dx} = -\frac{y}{x^2} \quad (x \neq 0)$$

$$\frac{dy}{y} = -\frac{dx}{x^2}$$

两边积分 $\ln|y| = \frac{1}{x} + C$

$$y = ce^{\frac{1}{x}} \quad (x \neq 0)$$

同时, $y=0$ 也是解

II). $x \sec y dx + (x+1) dy = 0$

$$\cos y dy = -\frac{x}{x+1} dx$$

两边积分 $\sin y = \ln|x| - x + C$

III). $(x^2+1) \frac{dy}{dx} = xy$

$$\frac{dy}{y} = \frac{x}{x^2+1} dx$$

两边积分 $\ln|y| = \frac{1}{2} \ln(x^2+1) + C$

$$y^2 = C(x^2+1)$$

IV). $yy' + e^{y^2+3x} = 0$

$$y \frac{dy}{dx} = -e^{y^2} \cdot e^{3x}$$

$$y e^{-y^2} dy = -e^{3x} dx$$

两边积分 $-\frac{1}{2} e^{-y^2} = -\frac{1}{3} e^{3x} + C$

$$3e^{-y^2} = 2e^{3x} + C$$

P4 习题 6 P5

$$I) \quad x \frac{dy}{dx} = y + x \sec \frac{y}{x}$$

$$\text{令 } y = ux \quad \frac{dy}{dx} = x \frac{du}{dx} + u$$

$$\text{又: } \frac{dy}{dx} = \frac{y}{x} + \sec \frac{y}{x} = u + \sec u$$

$$\sec u = x \frac{du}{dx}$$

$$\cos u du = \frac{dx}{x}$$

$$\text{两边积分} \quad \sin u = \ln|x| + C$$

$$e^{\sin u} = Cx$$

$$\text{解: } e^{\sin \frac{y}{x}} = Cx$$

$$II) \quad (x+y)dx + (y-x)dy = 0$$

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

$$\text{令 } y = ux \quad \frac{dy}{dx} = x \frac{du}{dx} + u$$

$$\frac{dy}{dx} = \frac{x(u+1)}{x(1-u)} = \frac{u+1}{1-u} = x \frac{du}{dx} + u$$

$$\frac{1-u}{1+u^2} du = \frac{dx}{x}$$

$$\text{两边积分} \quad \int \frac{1-u}{1+u^2} du - \int \frac{u}{1+u^2} du = \ln|x| + C$$

$$\arctan u - \frac{1}{2} \ln(1+u^2) = \ln|x| + C$$

$$\arctan\left(\frac{y}{x}\right) - \frac{1}{2} \ln\left(1 + \frac{y^2}{x^2}\right) = \ln|x| + C$$

$$\text{解: } \arctan\left(\frac{y}{x}\right) = \frac{1}{2} \ln(x^2 + y^2) + C$$

$$III) \quad xy' - y = xe^{\frac{y}{x}} = 0$$

$$\text{令 } y = ux \quad \frac{dy}{dx} = x \frac{du}{dx} + u$$

$$x \frac{dy}{dx} = y + xe^{\frac{y}{x}}$$

$$\frac{dy}{dx} = \frac{1}{x} (ux + xe^u)$$

$$e^u = \frac{du}{dx}$$

$$e^{-u} du = dx$$

两边积分 $-e^{-u} = \ln|x| + C_1$

解: $-e^{-\frac{y}{x}} = \ln(Cx)$

IV). $xy' \sin \frac{y}{x} - y \sin \frac{y}{x} + x = 0$

令 $y = ux$ $\frac{dy}{dx} = x \frac{du}{dx} + u$

$$xy' \sin u - ux \sin u + x = 0$$

$$\frac{du}{dx} = \frac{u \sin u - 1}{\sin u} = u - \frac{1}{\sin u}$$

$$-\frac{1}{\sin u} = x \frac{du}{dx}$$

$$-\sin u du = \frac{dx}{x}$$

两边积分 $\cos u = \ln|x| + C_1$

解: $\cos \frac{y}{x} = \ln(Cx)$

P5 习题 6 P6

I). $y' = \left(\frac{2}{x+y}\right)^2$

令 $x+y = u$ 则 $y = u - x$ $\frac{dy}{dx} = \frac{du}{dx} - 1$

$$\frac{du}{dx} - 1 = \frac{4}{u^2}$$

$$\frac{u^2}{4+u^2} du = dx$$

两边积分 $\int du - \int \frac{4}{4+u^2} du = x + C_1$

$$u - 2 \arctan \frac{u}{2} = x + C_1$$

$$x+y - 2 \arctan \frac{x+y}{2} = x + C_1$$

解: $y - 2 \arctan \frac{x+y}{2} + C = 0$

$$\text{II). } y' = \frac{2x+4y+3}{x+2y+1}$$

$$y' = \frac{2(x+2y)+3}{(x+2y)+1}$$

$$\text{令 } u = x+2y \rightarrow y = \frac{1}{2}(u-x) \quad \frac{dy}{dx} = \frac{1}{2}\left(\frac{du}{dx} - 1\right)$$

$$\frac{1}{2}\left(\frac{du}{dx} - 1\right) = \frac{2u+3}{u+1}$$

$$\frac{du}{dx} = \frac{4u+6}{u+1} + 1 = \frac{5u+7}{u+1}$$

$$\left(\frac{u+1}{5u+7}\right) du = dx$$

$$\text{两边积分 } \frac{1}{5}u - \frac{2}{25}\ln(5u+7) = x + C_1$$

$$\frac{1}{5}(x+2y) - \frac{2}{25}\ln(5x+10y+7) = x + C_1$$

$$\ln(5x+10y+7) = -10x + 5y - C_1$$

$$\text{解: } 5x+10y+7 = Ce^{-10x+5y}$$

$$\text{III). } y' = \frac{1}{(4x+y+1)(4xy)}$$

$$y' = \frac{1}{4xy} - \frac{1}{4xy+1}$$

$$\text{令 } u = 4x+y \rightarrow y = u-4x, \quad \frac{dy}{dx} = \frac{du}{dx} - 4$$

$$\frac{dy}{dx} = \frac{1}{u} - \frac{1}{u+1} = \frac{du}{dx} - 4$$

$$\frac{u(u+1)}{(2u+1)^2} du = dx$$

$$\text{两边积分 } \int \frac{1}{4} - \frac{\frac{1}{4}}{(2u+1)^2} du = \int dx + C_1$$

$$\frac{1}{4}u + \frac{1}{8(2u+1)} = x + C$$

$$2(4x+y) + \frac{1}{8x+2y+1} = 8x + C$$

$$\text{解: } 2y + \frac{1}{8x+2y+1} = C$$

$$\text{IV). } \frac{dy}{dx} = \frac{y-x+1}{y+x+5}$$

$$\begin{cases} y-x+1=0 \\ y+x+5=0 \end{cases} \Rightarrow \begin{cases} x=-2 \\ y=-3 \end{cases}$$

$$\text{令 } X = x+2 \quad Y = y+3$$

$$\frac{dy}{dx} = \frac{dY}{dX} = \frac{Y-X}{Y+X}$$

$$\text{令 } Y = uX \quad \text{则 } \frac{dY}{dX} = X \frac{du}{dX} + u$$

$$\frac{u-1}{u+1} = X \frac{du}{dX} + u$$

$$\frac{u-1}{u^2+1} du = -\frac{dx}{X}$$

$$\text{两边积分 } \frac{1}{2} \ln|u^2+1| + \arctan u = -\ln|X| + C_1$$

$$\ln\left(\frac{\sqrt{Y^2+X^2}}{X}\right) + \arctan \frac{Y}{X} = -\ln|X| + C_1$$

$$\text{解: } \frac{1}{2} \ln((y+3)^2 + (x+2)^2) + \arctan \frac{y+3}{x+2} = C$$

P6 习题6 P7.

$$I). \frac{dy}{dx} = y^2 + 1$$

$$\frac{dy}{y^2+1} = dx$$

$$\text{两边积分 } \arctan y = x + C$$

$$\text{代入 } y(1)=0 \quad 1+C=0 \rightarrow C=-1$$

$$\text{解: } \arctan y = x-1$$

$$II). \frac{dy}{dx} = \frac{xy+3x}{x^2+1}$$

$$\frac{dy}{y+3} = \frac{x dx}{x^2+1}$$

$$\text{两边积分 } \ln|y+3| = \frac{1}{2} \ln|x^2+1| + C_1$$

$$(y+3)^2 = C(x^2+1)$$

$$\text{代入 } y|x=2=2 \quad 25=5C \rightarrow C=5$$

$$\text{解: } (y+3)^2 = 5(x^2+1)$$

$$\text{IV). } y' = \frac{1}{x-y} + 1$$

$$y' = \frac{x-y+1}{x-y}$$

$$\text{令 } u = x-y \rightarrow y = -u+x \quad \frac{dy}{dx} = -\frac{du}{dx} + 1$$

$$\frac{dy}{dx} = \frac{u+1}{u}$$

$$\text{故 } 1 + \frac{1}{u} = -\frac{du}{dx} + 1$$

$$-u du = dx$$

$$\text{两边积分} \quad -\frac{u^2}{2} = x + C_1$$

$$x + \frac{(x-y)^2}{2} = C$$

$$\text{代入 } y(0)=1 \quad \frac{(-1)^2}{2} = \frac{1}{2} = C$$

$$\text{解: } x + \frac{(x-y)^2}{2} = \frac{1}{2}$$

$$\text{IV). } y' = \frac{x}{y} + \frac{y}{x}$$

$$\text{令 } y = ux, \text{ 则 } \frac{dy}{dx} = x \frac{du}{dx} + u$$

$$\frac{dy}{dx} = \frac{1}{u} + u$$

$$\text{故 } \frac{1}{u} = x \frac{du}{dx}$$

$$u du = \frac{dx}{x}$$

$$y^2 = 2x^2 (\ln|x| + C)$$

$$\text{两边积分} \quad \frac{1}{2}u^2 = \ln|x| + C_1$$

$$e^{\frac{u^2}{2}} = Cx$$

$$\text{代入 } y(1)=2 \quad e^2 = C$$

$$\text{解: } e^{\frac{y^2}{2x^2}} = e^2 x$$

P7 习题 6 P8

$$\text{I) } y' - 3y = e^x$$

$$p(x) = -3 \quad Q(x) = e^x$$

$$\begin{aligned} \text{Ans: } y &= e^{\int p(x) dx} \left(\int Q(x) e^{\int p(x) dx} dx + C_1 \right) \\ &= e^{3x} \left(\int e^{-2x} dx + C_1 \right) \\ &= e^{3x} \left(-\frac{1}{2} e^{-2x} + C \right) \\ &= C e^{3x} - \frac{1}{2} e^x \end{aligned}$$

$$\text{II). } y' + 4y = x$$

$$p(x) = 4 \quad Q(x) = x$$

$$\begin{aligned} \text{Ans: } y &= e^{\int p(x) dx} \left(\int Q(x) e^{\int p(x) dx} dx + C_1 \right) \\ &= e^{-4x} \left(\int x e^{4x} dx + C_1 \right) \\ &= e^{-4x} \left[\frac{1}{4} (x e^{4x} - \int e^{4x} dx) + C_1 \right] \\ &= \frac{1}{4} e^{-4x} \left(x e^{4x} - \frac{1}{4} e^{4x} + C \right) \\ &= \frac{C}{4} e^{-4x} + \frac{x}{4} - \frac{1}{16} \end{aligned}$$

$$\text{III) } y' - 2xy = x$$

$$p(x) = -2x \quad Q(x) = x$$

$$\begin{aligned} \text{Ans: } y &= e^{\int p(x) dx} \left(\int Q(x) e^{\int p(x) dx} dx + C_1 \right) \\ &= e^{x^2} \left(\int x e^{-x^2} dx + C_1 \right) \\ &= e^{x^2} \left(-\frac{1}{2} e^{-x^2} + C \right) \\ &= C e^{x^2} - \frac{1}{2} \end{aligned}$$

$$\text{IV). } \frac{dy}{dx} + y \cos x = e^{-\sin x}$$

$$p(x) = \cos x \quad Q(x) = e^{-\sin x}$$

$$\begin{aligned} \text{Ans: } y &= e^{\int p(x) dx} \left(\int Q(x) e^{\int p(x) dx} dx + C_1 \right) \\ &= e^{-\sin x} \left(\int e^{-\sin x} e^{\sin x} dx + C_1 \right) \end{aligned}$$

$$= e^{-\sin x} (\lambda + C)$$

$$V). y' \cos x = y \sin x + \sin 2x$$

$$x \in (-\frac{\pi}{2}, \frac{\pi}{2}) \Rightarrow \cos x \neq 0$$

$$y' - \tan x y = 2 \sin x$$

$$p(x) = -\tan x, \quad Q(x) = 2 \sin x$$

$$\text{解: } y = e^{\int p(x) dx} \left(\int Q(x) e^{\int p(x) dx} dx + C_1 \right)$$

$$= \sec x \left(\int 2 \sin x \cos x dx + C_1 \right)$$

$$= \sec x \left(-\frac{1}{2} \cos 2x + C_1 \right)$$

$$= C \sec x - \cos x$$

$$VI). dx + (x + y^2) dy = 0$$

$$\frac{dx}{dy} + x + y^2 = 0$$

$$\frac{dx}{dy} + x = -y^2$$

$$p(y) = 1, \quad Q(y) = -y^2$$

$$\text{解: } x = e^{\int p(y) dy} \left(\int Q(y) e^{\int p(y) dy} dy + C_1 \right)$$

$$= e^{-y} \left(\int -y^2 e^y dy + C_1 \right) - 2y$$

$$= e^{-y} \left(-y^2 e^y + \int 2y e^y dy + C_1 \right)$$

$$= e^{-y} \left(-y^2 e^y + 2y e^y - \int 2e^y dy + C_1 \right)$$

$$= e^{-y} \left(-y^2 e^y + 2y e^y - 2e^y + C_1 \right)$$

$$= C e^{-y} - y^2 + 2y - 2$$

P8 习题 P9

$$I) y' + y = x + e^x$$

$$p(x)=1 \quad Q(x)=x e^x$$

$$\text{解: } y = e^{\int -p(x)dx} \left(\int Q(x) e^{\int p(x)dx} dx + C_1 \right)$$

$$= e^{-x} \left(\int (x+e^x) e^x dx + C_1 \right)$$

$$= e^{-x} \left(\int x e^x dx + \frac{1}{2} e^{2x} + C_1 \right)$$

$$= e^{-x} \left(x e^x - e^x + \frac{1}{2} e^{2x} + C \right)$$

$$= C e^{-x} + \frac{1}{2} e^x + x - 1$$

$$\text{代入 } y(0)=0 \quad C + \frac{1}{2} - 1 = 0 \rightarrow C = \frac{1}{2}$$

$$y = \frac{e^x e^x}{2} + x - 1 = \cosh x + x - 1$$

$$\text{II) } xy' - 3y = x^2$$

由于 $x \neq 0$,

$$y' - \frac{3}{x}y = x$$

$$p(x) = -\frac{3}{x}, \quad Q(x) = x$$

$$\text{解: } y = e^{\int p(x)dx} \left(\int Q(x) e^{\int p(x)dx} dx + C_1 \right)$$

$$= x^3 \left(\int x \cdot \frac{1}{x^3} dx + C_1 \right)$$

$$= x^3 \left(-\frac{1}{x} + C \right)$$

$$= x^2 (Cx - 1)$$

$$\text{代入 } y(1)=0 \quad C - 1 = 0 \rightarrow C = 1$$

$$y = x^2 (x - 1)$$

$$\text{III) } t \frac{dy}{dt} = -x + \sin t$$

$$x + \frac{1}{t}x = \frac{\sin t}{t}$$

$$p(t) = \frac{1}{t} \quad Q(t) = \frac{\sin t}{t}$$

$$\text{解: } x = e^{\int p(t)dt} \left(\int Q(t) e^{\int p(t)dt} dt + C_1 \right)$$

$$= \frac{1}{t} (\int \sin t dt + C_1)$$

$$= -\frac{1}{t} (\cos t + C)$$

$$= -\frac{C + \cos t}{t}$$

$$\text{代入 } x(\pi) = 1 \quad 1 = -\frac{C-1}{\pi} \rightarrow C = 1-\pi$$

$$x = \frac{\pi - 1 - \cos t}{t}$$

$$\text{IV) } y' + y \cot x = 5e^{\cos x}$$

$$p(x) = \cot x \quad Q(x) = 5e^{\cos x}$$

$$\text{解: } y = e^{\int -p(x) dx} (\int Q(x) e^{\int p(x) dx} dx + C_1)$$

$$= \frac{1}{\sin x} (\int 5 \sin x e^{\cos x} dx + C_1)$$

$$= \frac{1}{\sin x} (-5e^{\cos x} + C)$$

$$= \frac{C - 5e^{\cos x}}{\sin x}$$

$$\text{代入 } y\left(\frac{\pi}{2}\right) = -4 \quad -4 = \frac{C-5}{1} \rightarrow C = 1$$

$$y = \frac{1 - 5e^{\cos x}}{\sin x}$$

P9 习题 P10

$$\text{I) } y' - 3xy = xy^2$$

$$\text{令 } z = y^{-1} \text{ 则 } z' = -y^{-2}y', \quad y' = -y^2z' = -z^{-2}z'$$

$$-z^{-2}z' - 3xz^{-1} = xz^{-2}$$

$$z' + 3xz = -x$$

$$p(x) = 3x \quad Q(x) = -x$$

$$\text{解: } z = e^{\int -p(x) dx} (\int Q(x) e^{\int p(x) dx} dx + C_1)$$

$$= e^{-\frac{3}{2}x^2} (\int -xe^{\frac{3}{2}x^2} dx + C_1)$$

$$= e^{-\frac{3}{2}x^2} \left(-\frac{1}{3} e^{\frac{3}{2}x^2} + C \right)$$

$$= C e^{-\frac{3}{2}x^2} - \frac{1}{3}$$

$$\text{代入 } y = z^{-1}$$

$$y^{-1} = C e^{-\frac{3}{2}x^2} - \frac{1}{3}$$

$$\text{II). } x \frac{dy}{dx} + 2y = \frac{y^3}{x}$$

$$\frac{dy}{dx} + \frac{2}{x}y = \frac{1}{x^2}y^3$$

$$\text{令 } z = y^{-2} \text{ 则 } z' = -2y^{-3}y' \quad y' = -\frac{1}{2}y^3z'$$

$$-\frac{1}{2}y^3z' + \frac{2}{x}y = \frac{1}{x^2}y^3$$

$$-\frac{1}{2}z' + \frac{2}{x}z = \frac{1}{x^2}$$

$$z' - \frac{4}{x}z = -\frac{2}{x^2}$$

$$p(x) = -\frac{4}{x} \quad Q(x) = -\frac{2}{x^2}$$

$$\text{解: } z = e^{\int p(x)dx} \left(\int Q(x) e^{\int p(x)dx} dx + C_1 \right)$$

$$= x^4 \left(\int 2x^{-6} dx + C_1 \right)$$

$$= x^4 \left(\frac{2}{5} x^{-5} + C \right)$$

$$y^{-2} = C x^4 + \frac{2}{5x}$$

$$\text{IV) } y' + \frac{1}{x}y = x^{-\frac{1}{2}}y^{\frac{1}{2}}$$

$$\text{令 } z = y^{\frac{1}{2}}, \text{ 则 } z' = \frac{1}{2}y^{-\frac{1}{2}}y', \quad y' = 2y^{\frac{1}{2}}z'$$

$$2y^{\frac{1}{2}}z' + \frac{1}{x}y = x^{-\frac{1}{2}}y^{\frac{1}{2}}$$

$$2z' + \frac{1}{x}y^{\frac{1}{2}} = x^{-\frac{1}{2}}$$

$$z' + \frac{1}{2x}z = x^{-\frac{1}{2}}$$

$$p(x) = \frac{1}{2x} \quad Q(x) = x^{-\frac{1}{2}}$$

$$\text{解: } z = e^{\int p(x)dx} \left(\int Q(x) e^{\int p(x)dx} dx + C_1 \right)$$

$$= x^{-\frac{1}{2}} (\int dx + C_1)$$

$$= x^{-\frac{1}{2}} (x + C)$$

$$y^{\frac{1}{2}} = Cx^{-\frac{1}{2}} + x^{\frac{1}{2}}$$

P₁₀ 习题 P₁₁

$$f'(x) = \int_0^{2x} f(\frac{t}{2}) dt + \ln 2$$

$$f'(x) = 2f(x)$$

$$f(x) = Ce^{2x}$$

$$\text{又由 } f(0) = \ln 2 = C \text{ 得 } f(x) = \ln 2 e^{2x}$$

P₁₁ 习题 P₁₂

$$\int_0^1 f(x) dx = \frac{1}{2} f(x) + 1$$

$$\text{令 } y = x \text{ 则 } dy = x dx$$

$$\int_0^x f(y) \frac{dy}{x} = \frac{1}{2} f(x) + 1$$

$$\int_0^x f(y) dy = \frac{x}{2} f(x) + x$$

$$f'(x) = \frac{1}{2} f(x) + \frac{x}{2} f'(x) + 1$$

$$f'(x) - \frac{1}{2} f(x) = -\frac{2}{x}$$

$$p(x) = -\frac{1}{x} \quad q(x) = -\frac{2}{x}$$

$$\text{解: } f(x) = e^{\int p(x) dx} (\int q(x) e^{\int p(x) dx} dx + C_1)$$

$$= x (\int -\frac{2}{x^2} dx + C_1)$$

$$= x (\frac{2}{x} + C)$$

$$= Cx + 2$$

P12 习题 6 P13

$$2 \int_0^x y(t) \sqrt{1+y'^2} dt = 2x + y^2(x)$$

$$2y(x) \sqrt{1+y'^2(x)} = 2 + 2y(x) y'(x)$$

$$y(x) \sqrt{1+y'^2(x)} = 1 + y(x) y'(x)$$

$$\text{令 } p = y'(x)$$

$$y \sqrt{1+p^2} = 1 + yp$$

$$y^2(1+p^2) = y^2 p^2 + 2yp + 1$$

$$y^2 = 2yp + 1$$

$$p = \frac{y^2 - 1}{2y}$$

$$\text{故 } \frac{dy}{dx} = \frac{y^2 - 1}{2y}$$

$$\frac{2y}{y^2 - 1} dy = dx$$

$$\text{两边积分 } \ln|y^2 - 1| = x + C_1$$

$$y^2 = ce^x + 1$$

$$\text{由 } y(0) = C + 1 = 0 \text{ 得 } C = -1$$

$$\text{故 } y = \sqrt{1 - e^x}$$

P13 习题 6 P14

$$\frac{dR}{dt} = -kR$$

$$\frac{dR}{R} = -k dt$$

$$\text{两边积分 } \ln R = -kt + C_1$$

$$R = ce^{-kt}$$

$$\text{代入 } R|_{t=0} = R_0 \quad R|_{t=1600} = \frac{1}{2} R_0$$

$$\begin{cases} R_0 = C \\ \frac{1}{2}R_0 = Ce^{-1600k} \end{cases} \Rightarrow \begin{cases} C = R_0 \\ k = \frac{\ln 2}{1600} \end{cases}$$

$$R = R_0 e^{-\frac{\ln 2}{1600} t}$$

当 $R = \frac{1}{10}R_0$ 时

$$e^{-\frac{\ln 2}{1600} t_0} = \frac{1}{10}$$

$$t_0 \approx 5315 \text{ year.}$$

$$t = t_0 - 1600 = 3715 \text{ year.}$$

P₁₄ 题 6 P₅

$$C: y - y_0 = y'|_{x=x_0}(x - x_0)$$

$$\text{过 } (0, -x_0 y'|_{x=x_0+y_0}) \text{ 和 } (-\frac{y_0}{y'|_{x=x_0}} + x_0, 0)$$

$$\text{且 } \begin{cases} 2x_0 = -\frac{y_0}{y'|_{x=x_0}} + x_0 \\ 2y_0 = -x_0 y'|_{x=x_0+y_0} + y_0 \end{cases} \Rightarrow y_0 = -x_0 y'|_{x=x_0}$$

$$y = -x \frac{dy}{dx}$$

$$\frac{dy}{y} = -\frac{dx}{x}$$

$$\text{两边积分 } \ln|y| = -\ln|x| + C_1$$

$$\ln xy = C$$

$$\text{代入 (2,3) } \ln 6 = C$$

$$\text{曲线: } xy = 6$$

P₁₅ 题 6 P₉

设 t 时刻 CO 含量为 $y(t)$ (cm^3)

注入 CO 速率为 $0.05 \times 6\% = 3 \times 10^{-4} \text{ m}^3/\text{min}$, $c(t) = \frac{y(t)}{500 + (0.05 - 0.06)t}$

故排出 CO 的速率为 $\frac{0.06 y(t)}{500 - 0.01t}$

$$\frac{dy}{dt} = 3 \times 10^{-4} - \frac{0.06 y(t)}{500 - 0.01t}$$

$$p(t) = \frac{0.06}{500 - 0.01t} \quad Q(t) = 3 \times 10^{-4}$$

解: $y = e^{\int p(t) dt} \left(\int Q(t) e^{\int p(t) dt} dt + C_1 \right)$

$$= (-0.01t + 500)^6 \left(\int 3 \times 10^{-4} (-0.01t + 500)^{-6} dt + C_1 \right)$$

$$= (-0.01t + 500)^6 (0.006 (-0.01t + 500)^{-5} + C_1)$$

$$500^6 (0.006 (500)^{-5} + C_1) = 0 \Rightarrow C_1 = -0.006 (500)^{-5}$$

当 $y = 500 \times 0.1\% = 0.5 \text{ m}^3$ 时, $t \approx 186.5 \text{ (min)}$

P16 习题 6 P20.

I). $y'' = x + \cos x$

$$y' = \frac{1}{2}x^2 + \sin x + C_1$$

$$y = \frac{1}{6}x^3 - \cos x + C_1 x + C_2$$

II). $y'' = \frac{1}{1+x^2}$

$$y' = \arctan x + C_1$$

$$y = x \arctan x - \int \frac{x}{1+x^2} dx + C_1 x + C_2$$

$$= x \arctan x - \frac{1}{2} \ln(1+x^2) + C_1 x + C_2$$

III). $4xy'' - y'' - 4y' = 0$

$$\text{令 } y' = p$$

$$4x p' - p' - 4p = 0$$

$$(4x-1) \frac{dp}{dx} = 4p$$

$$\frac{dp}{4p} = \frac{dx}{4x-1}$$

两边积分得 $\ln|p| = \ln|x-\frac{1}{4}| + C$

$$\frac{dy}{dx} = p = C_1(x-\frac{1}{4})$$

两边积分得 $y = C_2(x-\frac{1}{4})^2 + C_3$

IV) $yy'' - (y')^2 = 0$

$$(\frac{y'}{y})' = \frac{yy'' - (y')^2}{y^2} = 0$$

$$\frac{y'}{y} = C_1$$

$$\frac{dy}{dx} = C_1 y$$

$$\frac{dy}{y} = C_1 dx$$

两边积分 $\ln y = C_1 x + C$

$$y = C_2 e^{C_1 x}$$

V) $y'' = (y')^2 + y'$

令 $y' = p(y)$ 则 $y'' = p \frac{dp}{dy}$

$$p \frac{dp}{dy} = p^2 + p$$

$$\frac{dp}{dy} = p^2 + 1$$

$$\frac{dp}{p^2+1} = dy$$

两边积分 $\arctan p = y + C$

$$\frac{dy}{dx} = p = \tan(y+C)$$

$$\frac{dy}{\tan(y+C)} = dx$$

两边积分 $\ln \sin(y+C) = x + C_1$

$$\sin(y+C) = C_1 e^x$$

$$y = \arcsin(C_1 e^x) + C_1$$

$$\text{VI) } y''' = y''$$

$$\text{令 } p = y''$$

$$\text{则 } p' = p$$

$$p = C_1 e^x$$

$$y'' = C_1 e^x$$

$$\text{两边积分 } y' = C_1 e^x + C_2$$

$$y = C_1 e^x + C_2 x + C_3$$

P17 习题 6 P21

$$\text{I) } y'' - y'^2 = 0$$

$$\text{令 } p(y) = y'$$

$$p' - p^2 = 0$$

$$\frac{dp}{p^2} = dx$$

$$\text{两边积分 } -\frac{1}{p} = x + C_1$$

$$y' = p = -\frac{1}{x + C_1}$$

$$\text{两边积分 } y = -\ln(x + C_1) + C_2$$

$$\text{代入 } y(0) = 0, y'(0) = -1$$

$$\begin{cases} -\ln C_1 + C_2 = 0 \\ -\frac{1}{C_1} = -1 \end{cases} \rightarrow \begin{cases} C_1 = 1 \\ C_2 = 0 \end{cases}$$

$$\text{解: } y = -\ln(x + 1)$$

$$\text{II) } y^3 y'' + 1 = 0$$

$$\text{令 } y' = p(y) \text{ 则 } y'' = p \frac{dp}{dy}$$

$$y^3 \cdot p \frac{dp}{dy} + 1 = 0$$

$$p dp = -\frac{1}{y^3} dy$$

两边积分 $\frac{1}{2} p^2 = \frac{1}{2y^2} + C$

$$p^2 = \frac{1}{y^2} + C_1$$

$$y' = \frac{\sqrt{1+C_1 y^2}}{y}$$

$$\frac{y dy}{\sqrt{1+C_1 y^2}} = dx$$

两边积分 $\frac{1}{C_1} \sqrt{1+C_1 y^2} = x + C_2$

代入 $y(1)=1, y'(1)=0$

$$\begin{cases} \frac{1}{C_1} \sqrt{1+C_1} = 1 + C_2 \\ 0 = \frac{\sqrt{1+C_1}}{1} \end{cases} \rightarrow \begin{cases} C_1 = -1 \\ C_2 = -1 \end{cases}$$

$$-\sqrt{1-y^2} = x-1$$

解: $y = \sqrt{2x-x^2}$

四. $yy'' = (y')^2 - (y')^3$

令 $p = y'$ 则 $y'' = p \frac{dp}{dy}$

$$y p \frac{dp}{dy} = p^2 - p^3$$

$$y \frac{dp}{dy} = p - p^2$$

$$\frac{dp}{p(1-p)} = \frac{dy}{y}$$

两边积分 $\ln \left| \frac{p}{1-p} \right| = \ln |y| + C$

$$\frac{y}{1-y} = \frac{p}{1-p} = C_1 y$$

$$\frac{dy}{dx} = \frac{C_1 y}{C_1 y + 1}$$

$$\frac{C_1 y + 1}{C_1 y} dy = dx$$

两边积分 $y + \frac{1}{C_1} \ln y = x + C_2$

代入 $y(1)=1, y'(1)=1$

$$\begin{cases} 1 = 1 + C_2 \\ \frac{-1}{1+1} = C_1 \end{cases} \Rightarrow \begin{cases} C_1 = -\frac{1}{2} \\ C_2 = 0 \end{cases}$$

解: $y = e^{\frac{x-1}{2}}$

IV. $xy'' + (xy')^2 - y' = 0$

令 $y' = p(x)$ 则 $y'' = p'(x)$

$$xp' + xp^2 - p = 0$$

$$p' - \frac{1}{x}p = -p^2$$

令 $z = p^{-1}$ 则 $z' = -p^{-2}p'$ $p' = -p^2z'$

$$-p^2z' - \frac{1}{x}p = -p^2$$

$$z' + \frac{1}{x}p^{-1} = 1$$

$$z' + \frac{1}{x}z = 1$$

$p(x) = \frac{1}{x}, q(x) = 1$

故: $z = e^{\int -\frac{1}{x}dx} (\int q(x)e^{\int \frac{1}{x}dx} dx + C_1)$

$$= \frac{1}{x} (\frac{1}{2}x^2 + C)$$

$$p = \frac{x}{\frac{1}{2}x^2 + C} = \frac{dy}{dx}$$

$$dy = \frac{2x}{x^2 + C_1} dx$$

两边积分 $y = \ln(x^2 + C_1) + C_2$

代入 $y(2)=2, y'(2)=1$

$$\begin{cases} 2 = \ln(4 + C_1) + C_2 \\ 1 = \frac{4}{4 + C_1} \end{cases} \Rightarrow \begin{cases} C_1 = 0 \\ C_2 = 2 - 2\ln 2 \end{cases}$$

解: $y = 2(\ln \frac{x}{2} + 1)$

P18 题6 P23

$$f = kv^2$$

$$m \frac{dv}{dt} = mg - kv^2$$

$$t \rightarrow +\infty, \frac{dv}{dt} = 0$$

$$mg = kv_m^2 \rightarrow k = \frac{mg}{v_m^2}$$

$$\frac{dv}{dt} = g - \frac{v^2}{v_m^2}$$

$$\frac{dv}{1 - \frac{v^2}{v_m^2}} = g dt$$

$$\text{两边积分} \quad \frac{v_m}{2} \ln \left(\frac{v_m + v}{v_m - v} \right) = gt + C$$

$$\text{代入 } v|_{t=0} = 0 \quad 0 = C$$

$$\text{故} \quad \ln \left(\frac{v_m + v}{v_m - v} \right) = \frac{2g}{v_m} t$$

$$\frac{v_m + v}{v_m - v} = e^{\frac{2g}{v_m} t}$$

$$v(t) = v_m \frac{e^{\frac{2g}{v_m} t} - 1}{e^{\frac{2g}{v_m} t} + 1}$$

$$s(t) = \int_0^t v(\tau) d\tau = \frac{v_m^2}{g} \ln \left(\cosh \left(\frac{g}{v_m} t \right) \right) = \frac{75^2}{g} \ln \left(\cosh \left(\frac{g}{75} t \right) \right)$$

P19 题6 P1

$$I) y = x^3 - Cx$$

$$y' = 3x^2 - C \rightarrow C = 3x^2 - y'$$

$$\text{又 } C = \frac{y - x^3}{x}$$

$$\frac{y - x^3}{x} = 3x^2 - y'$$

$$\text{得 } xy' - y = 2x^3$$

$$I) (x - C)^2 + y^2 = 1$$

$$2(x - C) + 2yy' = 0 \rightarrow C = yy' + x$$

$$\text{又 } C = x - \sqrt{1-y}$$

$$x - \sqrt{1-y} = yy' + x$$

$$\text{得 } y^2 y'^2 + y = 1$$

$$\text{III. } y = (C_1 + C_2 x) e^{-x}$$

$$y' = C_2 e^{-x} - (C_1 + C_2 x) e^{-x}$$

$$y'' = e^{-x} (C_1 - 2C_2 + C_2 x)$$

$$\text{有 } y'' + 2y' + y = e^{-x} (C_1 - 2C_2 + C_2 x + 2C_2 - 2C_1 - 2C_2 x + C_1 + C_2 x) = 0$$

$$\text{得 } y'' + 2y' + y = 0$$

$$\text{IV) } y = Cx + C^2$$

$$y' = C$$

$$y = y'x + y'^2$$

$$\text{得 } y'^2 + y'x - y = 0$$

P₂₀ 补充题6 P₂

$$\text{I) } xy' + y = y (\ln x + \ln y)$$

$$(xy)' = y (\ln xy)$$

$$\text{令 } u = xy$$

$$\frac{du}{dx} = \frac{u}{x} \ln u$$

$$\frac{du}{u \ln u} = \frac{dx}{x}$$

$$\text{两边积分 } \ln |\ln u| = \ln |x| + C_1$$

$$\ln u = Cx$$

$$\text{解: } xy = e^{Cx}$$

$$II) \frac{dy}{dx} = \frac{4x^3y}{x^4+y^2}$$

$$\frac{dx}{dy} = \frac{x^4+y^2}{4x^3y} = \frac{x}{4y} + \frac{y}{4x^3}$$

$$x' - \frac{1}{4y}x = \frac{y}{4}x^{-3}$$

$$\text{令 } z = x^4, \text{ 则 } z' = 4x^3x' \quad x' = \frac{1}{4}x^3z'$$

$$\frac{1}{4}x^3z' - \frac{1}{4y}x = \frac{y}{4}x^{-3}$$

$$z' - \frac{1}{y}z = y$$

$$p(y) = -\frac{1}{y}, \quad Q(y) = y$$

$$z = e^{\int p(y)dy} \left(\int Q(y) e^{\int p(y)dy} dy + C \right)$$

$$x^4 = y(y+C) = Cy + y^2$$

$$\text{解: } x^4 = Cy + y^2$$

$$III). \quad (y^2 - bx)y' + 2y = 0$$

$$y' = \frac{2y}{bx - y^2}$$

$$\frac{dx}{dy} = \frac{bx - y^2}{2y} = \frac{b}{2}x - \frac{1}{2}y$$

$$x' - \frac{b}{2}x = -\frac{1}{2}y$$

$$p(y) = -\frac{b}{2}, \quad Q(y) = -\frac{1}{2}y$$

$$\text{解: } x = e^{\int p(y)dy} \left(\int Q(y) e^{\int p(y)dy} dy + C_1 \right)$$

$$= y^3 \left(\int -\frac{1}{2}y \cdot y^{-3} dy + C_1 \right)$$

$$= y^3 \left(\frac{1}{2y} + C \right)$$

$$= Cy^3 + \frac{1}{2}y^2$$

$$IV) \quad xy' - y = y^2 \ln x$$

$$y' - \frac{1}{x}y = \ln x y^2$$

$$\text{令 } z = y^{-1}, \text{ 则 } z' = -y^{-2}y', \quad y' = -y^2z'$$

$$-y^2 z' - \frac{1}{x} y = \ln x y^2$$

$$z' + \frac{1}{x} z = -\ln x$$

$$p(x) = \frac{1}{x} \quad Q(x) = -\ln x$$

$$z = e^{-\int p(x) dx} \left(\int Q(x) e^{\int p(x) dx} dx + C_1 \right)$$

$$= \frac{1}{x} \left(\int -x \ln x dx + C_1 \right)$$

$$y' = \frac{1}{x} \left(-\frac{1}{2} x^2 \ln x + \frac{1}{4} x^2 + C \right)$$

$$y' = \frac{C}{x} - \frac{1}{2} x \ln x + \frac{1}{4} x$$

$$v). \quad x(e^y - y') = 2$$

$$y' = e^y - \frac{2}{x}$$

$$\text{令 } u = e^y \quad \text{则 } u' = e^y \cdot y'$$

$$\frac{u'}{u} = u - \frac{2}{x}$$

$$u' + \frac{2}{x} u = u^2$$

$$\text{令 } z = u^{-1} \quad \text{则 } z' = -u^{-2} u', \quad u' = -u^2 z'$$

$$-u^2 z' + \frac{2}{x} u = u^2$$

$$z' - \frac{2}{x} z = -1$$

$$p(x) = -\frac{2}{x} \quad Q(x) = -1$$

$$z = e^{\int p(x) dx} \left(\int Q(x) e^{\int p(x) dx} dx + C_1 \right)$$

$$= x^2 \left(\int -\frac{1}{x^2} dx + C_1 \right)$$

$$= x \left(\frac{1}{x} + C \right)$$

$$\text{解: } e^{-y} = Cx^2 + 1$$

$$vi). \quad y' \tan y + \frac{1}{x} = e^x \cos y$$

$$y' \sec y \tan y + \frac{1}{x} \sec y = e^x$$

$$\text{令 } u = \sec y \quad \text{则 } u' = y' \sec y \tan y$$

$$u' + \frac{1}{x}u = e^x$$

$$p(x) = \frac{1}{x}, \quad Q(x) = e^x$$

$$u = e^{\int -p(x)dx} \left(\int Q(x) e^{\int p(x)dx} dx + C \right)$$

$$= \frac{1}{x} \left(\int x e^x dx + C \right)$$

$$\text{解: } \sec y = \frac{1}{x} (x e^x - e^x + C)$$

思考题

P₁

$$\text{I) } (P_{1.10}) \quad \frac{dT}{dt} = -k(T - T_e)$$

$$\frac{dT}{T - T_e} = -k dt$$

$$\text{两边积分 } \ln(T - T_e) = -kt + C_1$$

$$T = C e^{-kt} + T_e$$

$$\text{代入 } T|_{t=0} = 37 \quad T|_{t=2} = 35$$

$$\begin{cases} 17 = C \\ 15 = C e^{-2k} \end{cases} \Rightarrow \begin{cases} k = \frac{1}{2} \ln\left(\frac{17}{15}\right) \\ C = 17 \end{cases}$$

$$\text{故 } T = T_e + 17 e^{-\frac{1}{2} \ln\left(\frac{17}{15}\right) t}$$

$$\text{II) } T_e = T_0 - \frac{1}{2}t \quad (t \leq 0)$$

$$\frac{dT}{dt} = -k(T - T_0 + \frac{1}{2}t)$$

$$\frac{dT}{dt} + kT = T_0 k - \frac{1}{2}kt$$

$$p(t) = k \quad q(t) = 20k - \frac{1}{2}kt$$

$$T = e^{\int -p(t) dt} \left(\int q(t) e^{\int p(t) dt} dt + C_1 \right)$$

$$= e^{-kt} \left[\int (20k - \frac{1}{2}kt) e^{kt} dt + C_1 \right]$$

$$= e^{-kt} \left(20e^{kt} - \frac{1}{2}te^{kt} + \frac{1}{2k}e^{kt} + C \right)$$

$$= Ce^{-kt} - \frac{1}{2}t + \frac{1}{2k} + 20$$

$$\text{代入 } T|_{t=0} = 32 \rightarrow C = 12 - \frac{1}{2k}$$

$$\text{代入 } T|_{t=t_0} = 37$$

$$37 = \left(12 - \frac{1}{2k} \right) e^{-kt} - \frac{1}{2}t + \frac{1}{2k} + 20$$

$$\text{解得 } t \approx -6.20h$$

故更精确的死亡时间为 9:48 a.m.

P₂.

I). 初速度 v_0 , 质量 m , 阻力系数 k

$$f = kv$$

$$\frac{dv}{dt} = -\frac{f}{m} = -\frac{kv}{m}$$

$$\frac{dv}{v} = -\frac{k}{m} dt$$

$$\text{两边积分 } \ln|v| = -\frac{k}{m}t + C_1$$

$$v = Ce^{-\frac{k}{m}t}$$

$$\text{代入 } v|_{t=0} = v_0 \rightarrow C = v_0$$

$$v(t) = v_0 e^{-\frac{k}{m}t}$$

II). 最低阶高次项 v^2 , 系数 r .

$$f = kv + rv^2$$

$$\frac{dv}{dt} = -\frac{f}{m} = -\frac{kv+rv^2}{m} = -\frac{k}{m} \left(v + \frac{r}{k} v^2 \right)$$

$$\frac{dv}{v + \frac{r}{k} v^2} = -\frac{k}{m} dt$$

两边积分 $\ln|v| - \ln(rv+k) = -\frac{k}{m}t + C_1$

$$\frac{v}{rv+k} = C e^{-\frac{k}{m}t}$$

代入 $v|_{t=0} = v_0 \rightarrow C = \frac{v_0}{rv_0+k}$

$$v(t) = \frac{kv_0 e^{-\frac{k}{m}t}}{rv_0(1 - e^{-\frac{k}{m}t}) + k}$$